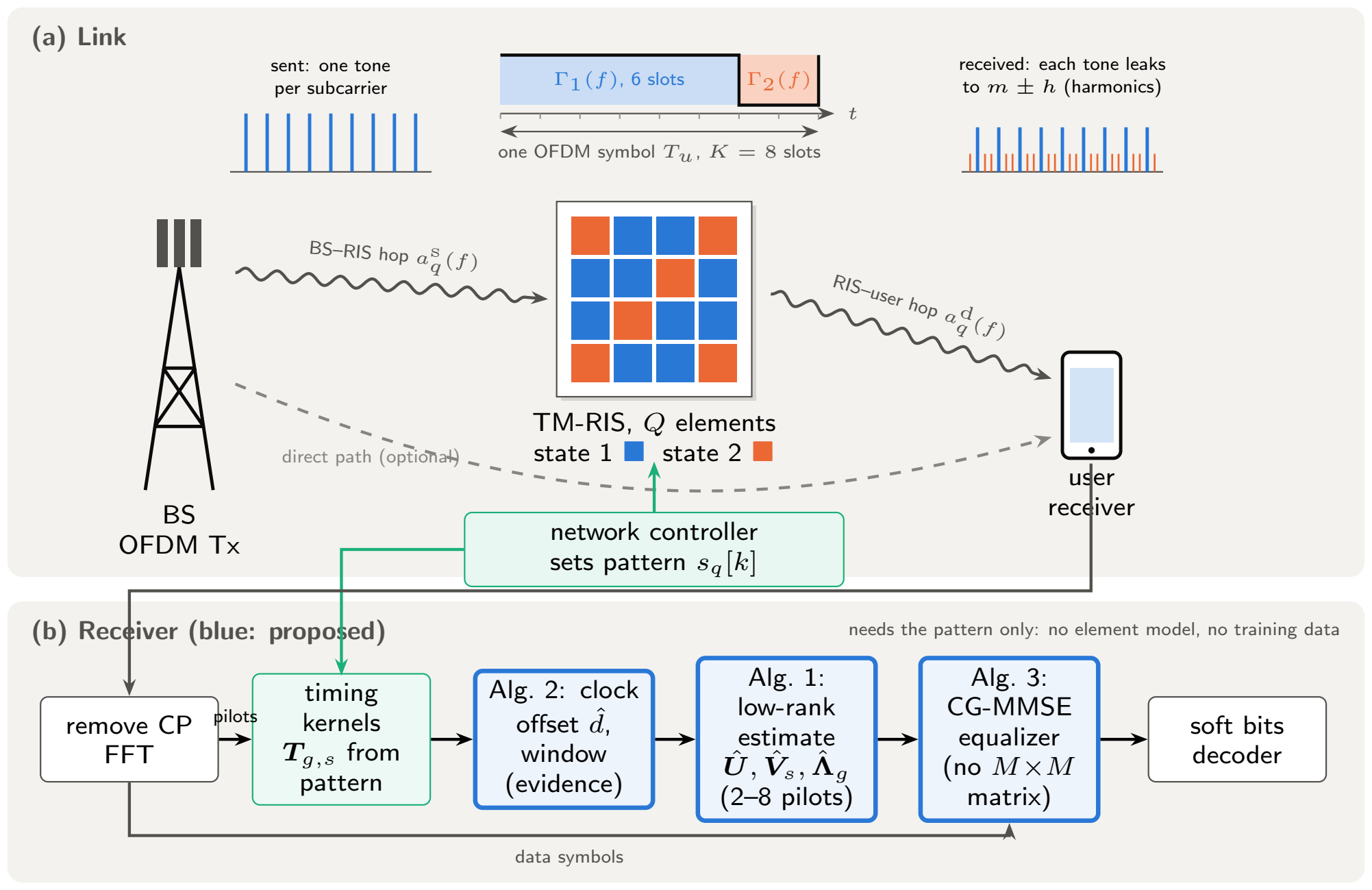

(a) Link
sent: one tone per subcarrier
$\Gamma_1(f)$, 6 slots
$\Gamma_2(f)$
t
one OFDM symbol T_u, $K = 8$ slots
received: each tone leaks to $m \pm h$ (harmonics)
BS–RIS hop $a_q^{\mathrm{s}}(f)$
RIS–user hop $a_q^{\mathrm{d}}(f)$
TM-RIS, Q elements
state 1
state 2
direct path (optional)
BS
OFDM Tx
user
receiver
network controller
sets pattern $s_q[k]$
(b) Receiver (blue: proposed)
needs the pattern only: no element model, no training data
remove CP
FFT
pilots
timing
kernels
$\boldsymbol{T}_{g,s}$ from
pattern
Alg. 2: clock
offset $\hat{d}$,
window
(evidence)
Alg. 1:
low-rank
estimate
$\hat{\boldsymbol{U}}, \hat{\boldsymbol{V}}_s, \hat{\boldsymbol{\Lambda}}_g$
(2–8 pilots)
Alg. 3:
CG-MMSE
equalizer
(no $M \times M$
matrix)
soft bits
decoder
data symbols